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Studierichting: Informatica
Oefeningenreeks 4A nr. 26.15

$$\begin{aligned}& \int (4x^2 + 2x + 1)(2x - 1)^{-3} dx \\& t = 2x - 1 \left(\Rightarrow dt = 2dx \Rightarrow dx = \frac{dt}{2} \Rightarrow x = \frac{t+1}{2} \right) \\& \int \left(4 \left(\frac{t+1}{2} \right)^2 + 2 \left(\frac{t+1}{2} \right) + 1 \right) t^{-3} \frac{dt}{2} \\& = \int \left(\frac{4(t+1)^2}{4} + \frac{2(t+1)}{2} + 1 \right) t^{-3} \frac{dt}{2} \\& = \int ((t+1)^2 + (t+1) + 1) t^{-3} \frac{dt}{2} \\& = \int (t^2 + 2t + 1 + t + 1 + 1) t^{-3} \frac{dt}{2} \\& = \int (t^2 + 3t + 3) t^{-3} \frac{dt}{2} \\& = \frac{1}{2} \int (t^2 + 3t + 3) t^{-3} dt \\& = \frac{1}{2} \int (t^{-1} + 3t^{-2} + 3t^{-3}) dt \\& = \frac{1}{2} \left(\ln |t| - 3t^{-1} - \frac{3}{2}t^{-2} \right) + C \\& = \frac{1}{2} \ln |t| - \frac{3}{2}t^{-1} - \frac{3}{4}t^{-2} + C \\& = \frac{1}{2} \ln |2x - 1| - \frac{3}{2(2x - 1)} - \frac{3}{4(2x - 1)^2} + C\end{aligned}$$

References